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TEC–802

B. TECH. (ECE) (EIGHTH SEMESTER)
MID SEMESTER
EXAMINATION, March, 2024

OPTICAL FIBER COMMUNICATION

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Define in detail the block diagram of an Optical Fiber Communication System. (CO2)

OR

(b) Define Critical Angle : (CO2)

Consider a multimode fiber that has a core refractive index of 1.48 and a core

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cladding index difference 2% ($\Delta = 0.020$). Find the :

(i) Numerical Aperture

(ii) Acceptance Angle

(iii) Critical Angle

2. (a) Describe the benefits and applications of optical fibre communication. (CO1)

OR

(b) Define the difference between Step Index Fiber and Graded Index Fiber. (CO1)

A step index fiber has a normalized frequency V = 29.9 at 13.50 nm wavelength, if the core radius is 35 μm. Calculate Numerical Aperture.

3. (a) Define Attenuation in optical fibre communication. Explain Snell's law. (CO2)

Consider the interference between a glass slab with $n_1 = 1.48$ and air for which $n_2 = 1.00$. What is the critical angle for light travelling in the glass ?

OR

(b) Show with neat diagrams the ray optics representation for the skew rays and meridional rays in a step index fiber. (CO1, CO2)

4. (a) A multiple step index fiber with a core diameter of 80 μm and a refractive index difference of 1.5% is operating at a wavelength of 0.85 μm. If the core refractive index is 1.48, estimate the normalized frequency for the fiber and the number of guided modes. (CO2)

OR

(b) Define Rayleigh Scattering Losses. (CO2)

5. (a) For single mode fibre, define the group and mode delay factors. Define Numerical Aperture. (CO1)

OR

(b) Define Step Index Fiber and Graded Index Fiber. Define the Ray theory. (CO1)